

1 Input data sources



Census & demographics

Population, age structure, households by parish



Mobility / immigration records

Device mobility, commuting, travel & migration flows



Reported cases

Confirmed cases by parish over time



Reported deaths

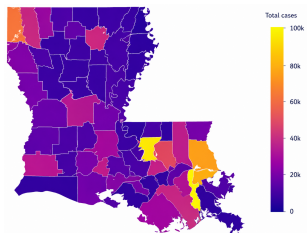
Deaths by parish over time



Geographic adjacency (parish neighbors)

Adjacency defines spatial structure & local interactions

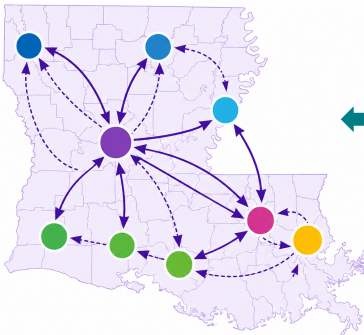
Observed cases (example snapshot)



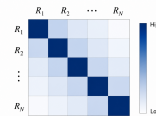
Louisiana parish-level confirmed cases

2 Population-immigration-informed regional network & inverse model

Louisiana parish regional network



Mobility prior / coupling matrix C



Encodes immigration / mobility strength between regions (row-normalized)

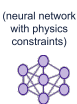
Geo-identifiable regional PINN (inverse model)

Epidemic dynamics per region i

$$\begin{aligned} \frac{dS_i}{dt} &= -\beta_i(t) \frac{S_i I_i}{N_i} + \Lambda_i(t) \\ \frac{dI_i}{dt} &= \beta_i(t) \frac{S_i I_i}{N_i} - \sigma E_i \\ \frac{dE_i}{dt} &= \sigma E_i - \gamma I_i \\ \frac{dR_i}{dt} &= \gamma I_i \end{aligned}$$

with immigration inflow $\Lambda_i(t)$

PINN



Learned quantities

- Transmission rate $\beta_i(t)$
- Importation rate α_i (baseline immigration)
- Coupling / mobility matrix C

→ Strong mobility / immigration

-----> Weak / long-range mobility

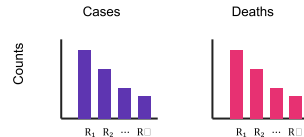
Learns epidemic dynamics + immigration + mobility structure across parishes and time

3 Aggregation to regional observations

Regional aggregation

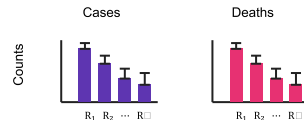
Sum model states to observation level (e.g., parishes or custom regions)

Model-predicted counts

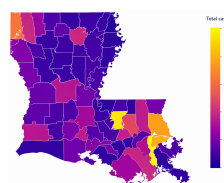


Observed counts

Reported observations



Example: parish-level snapshot



Observed counts used for training (and held-out evaluation)

4 Training objective & outputs

Training objective

$$\text{Total loss } \mathcal{L} = \lambda_p \mathcal{L}_{\text{phys}} + \lambda_d \mathcal{L}_{\text{data}}$$

Physics loss $\mathcal{L}_{\text{phys}}$

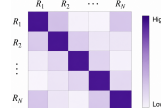
PINN residuals of SEIR + immigration & boundary conditions

Data loss $\mathcal{L}_{\text{data}}$

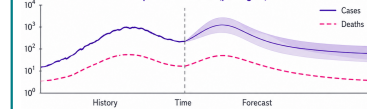
Mismatch between predicted and observed cases & deaths (Poisson/NB likelihood)

Outputs

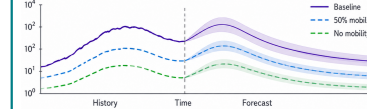
Recovered coupling (mobility) $\hat{C}$



Epidemic forecasts (per region)



Counterfactual scenarios



Optimization (training)

Minimize total loss using gradient-based methods (physics-informed, data-driven learning)